



Privacy in the «Big Data Era»

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Privacy under attack

- In 1996, 24% of Americans has “personally experienced a privacy invasion” (19% in 1978)
- Privacy vs. Convenience (trade off)?
 - In 1960s, some people said that polluting the environment is required for development

Right to privacy?

“The right to be left alone”

- Explicit right to privacy is given in constitutions, laws and international treaties
- People want to have privacy:
 - Objections to Intel's chip sets unique identification number
 - Objections to US National Database of Driver License Images
 - Objections to Facebook use of data
 - NSA scandals
 - Cambridge Analytica and Facebook scandal

Abundance of data

Increasing storage of personal data

- Credit card records
- Phone records
- Health care data
- E-mails
- Social accounts
- IoT devices
- Surveillance Cameras
- DNA data

Law of Unintended Consequences

- SSN was developed related to social security act
- SSN is even used for identifying students
- Cell phones are used for payments
- Lasers are used in DVD players
- What will be the DNA used for?

We live in the Age of Big Data

- “Big Data” refers to the acquisition and analysis of massive collections of information, collections so large that until recently the technology needed to analyze them did not exist
- And more:
 - Datafication
 - Massive amounts of unstructured “messy” data
 - Otherwise unnoticed patterns
 - Indiscriminate collection
 - Indefinite retention for unpredictable future uses

Threat: deanonymization

In 2006, Netflix released anonymized movie rating data for its Netflix Prize with date and value of movie ratings

- 100,000,000 (private) ratings from 500,000 users
- Competition to improve recommendations
 - i.e., if user X likes movies A,B,C, will also like D
- Anonymized: user name replaced by a number

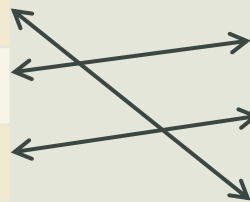


Threat: deanonymization

- Problem: can combine “private” ratings from Netflix with public reviews from IMDB to identify users in dataset
- May expose embarrassing info about members...



User	Movie	Rating
1234	Rocky II	3/5
1234	The Wizard	4/5
1234	The Dark Knight	5/5
...		
1234	Girls Gone Wild	5/5



User	Movie	Rating
dukefan	The Wizard	8/10
dukefan	The Dark Knight	10/10
dukefan	Rocky II	6/10
...		

Threat: deanonymization

Knowing 6-8 approximate movie ratings and dates is able to uniquely identify a record with over 90% probability

- Correlating with a set of 50 users from imdb.com yields two records

Netflix cancels second phase of the challenge

- **Lesson:** cannot always anonymize data simply by removing identifiers
- Vulnerable to aggregating data from multiple sources/networks
- Humans are predictable
 - E.g., try Rock-paper-scissors vs AI

AOL Data Release [New York Times]

- In August 2006, AOL Released search keywords of 650,000 users over a 3-month period.
 - User IDs are replaced by random numbers
 - 3 days later, pulled the data from public access

AOL searcher # 4417749

“landscapers in Lilburn, GA”
queries on last name “Arnold”
“homes sold in shadow lake
subdivision Gwinnett County, GA”
“num fingers”
“60 single men”
“dog that urinates on everything”

NYT

Thelman
Arnold, a 62
year old widow
who lives in
Liburn GA, has
three dogs,
frequently
searches her
friends' medical
ailments.



Examples of Re-identification Attempts

Examples	Health-specific and General Examples of Re-identification
AOL search data	Researchers were capable of revealing sensitive details of the participant's private lives, such as Social Security numbers, credit-card numbers, addresses etc. from the anonymized AOL Internet search data that contains health related searches as well
Chicago homicide database	A large percentage of individuals were re-identified easily by linking the Chicago homicide database with the social security death index
Netflix movie recommendations	Several individuals were re-identified from the publicly available anonymized Netflix movie recommendations database by linking their anonymized movie ratings with ratings in a publicly available Internet movie rating web site
Re-identification of the medical record	Massachusetts governor's sensitive medical records was re-identified by linking the anonymized data of the Group Insurance Commission, which purchases health insurance for state employees, with the voter list for Cambridge
Southern Illinoisan vs. The Department of Public Health	Individuals in a neuroblastoma data set from the Illinois cancer registry was re-identified with a very high accuracy
Canadian Adverse Event Database	An unfortunate death of a 26 year-old student by taking a particular drug was re-identified from the publicly released adverse drug reaction database of Health Canada

Most common causes of data breaches



COMPROMISED
CREDENTIALS (19%)

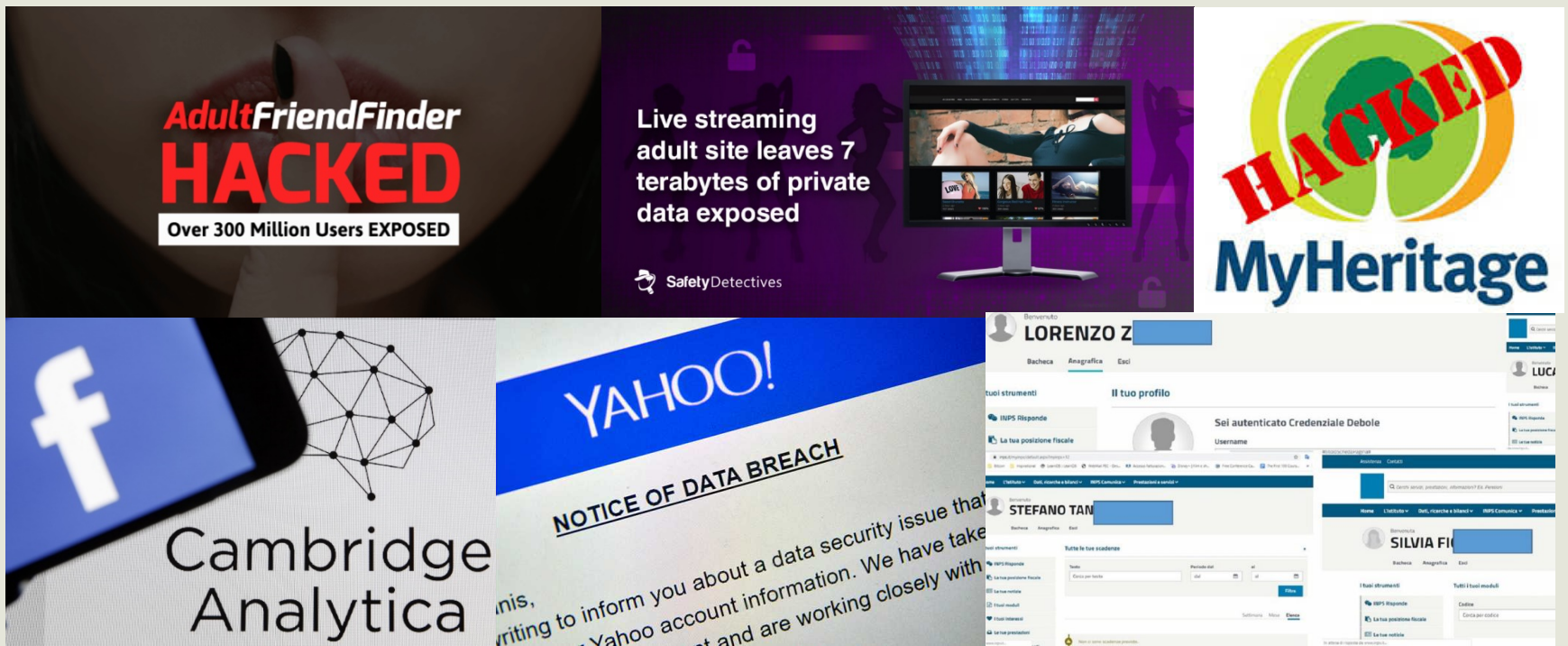


MISCONFIGURED
CLOUD SERVICES (19%)



SOFTWARE
VULNERABILITIES (16%)

Famous data breaches



Web 2.0 and social networks platforms reveal our (hidden) personal traits

Private traits and attributes are predictable from digital records of human behavior

Michal Kosinski^{a,1}, David Stillwell^a, and Thore Graepel^b

^aFree School Lane, The Psychometrics Centre, University of Cambridge, Cambridge CB2 3RQ United Kingdom; and ^bMicrosoft Research, Cambridge CB1 2FB, United Kingdom

Edited by Kenneth Wachter, University of California, Berkeley, CA, and approved February 12, 2013 (received for review October 29, 2012)

We show that easily accessible digital records of behavior, Facebook Likes, can be used to automatically and accurately predict a range of highly sensitive personal attributes including: sexual orienta-

browsing logs (11–15). Similarly, it has been shown that personality can be predicted based on the contents of personal Web sites (16), music collections (17), properties of Facebook or Twitter profiles

February 12, 2013 on Social Sciences

www.pnas.org/cgi/doi/10.1073/pnas.1218772110

The data analysis flow

1

Users' Facebook Likes

55,814 Likes

	art	cnn.com	(...)	BMW
User 1	1	1	...	0
User 2	0	1	...	1
User 3	1	0	...	0
(...)	...	...	...	...
User n	1	1	...	0

58,466 Users

User – Like Matrix
(10M User-Like pairs)

2

Singular Value Decomposition

100 Components

	Comp ₁	Comp ₂	(...)	Comp ₁₀₀
User 1	1.5	.7	...	-.9
User 2	.3	-.4	...	-.2
User 3	-.6	.1	...	4.7
(...)	...	...	...	...
User n	1.2	1	...	-.6

58,466 Users

User – Components Matrix

3

Prediction Model

Using Logistic or Linear Regression
(with 10-fold cross validation)

$$\text{e.g. } \text{age} = \alpha + \beta_1 C_1 + \dots + \beta_n C_{100}$$

Predicted variables

Facebook profile: age, gender, political and religious views, relationship status, proxy for sexual orientation, social network size and density

Profile picture: ethnicity

Survey / test results: BIG5 Personality, intelligence, satisfaction with life, substance use, parents together?

Personal history

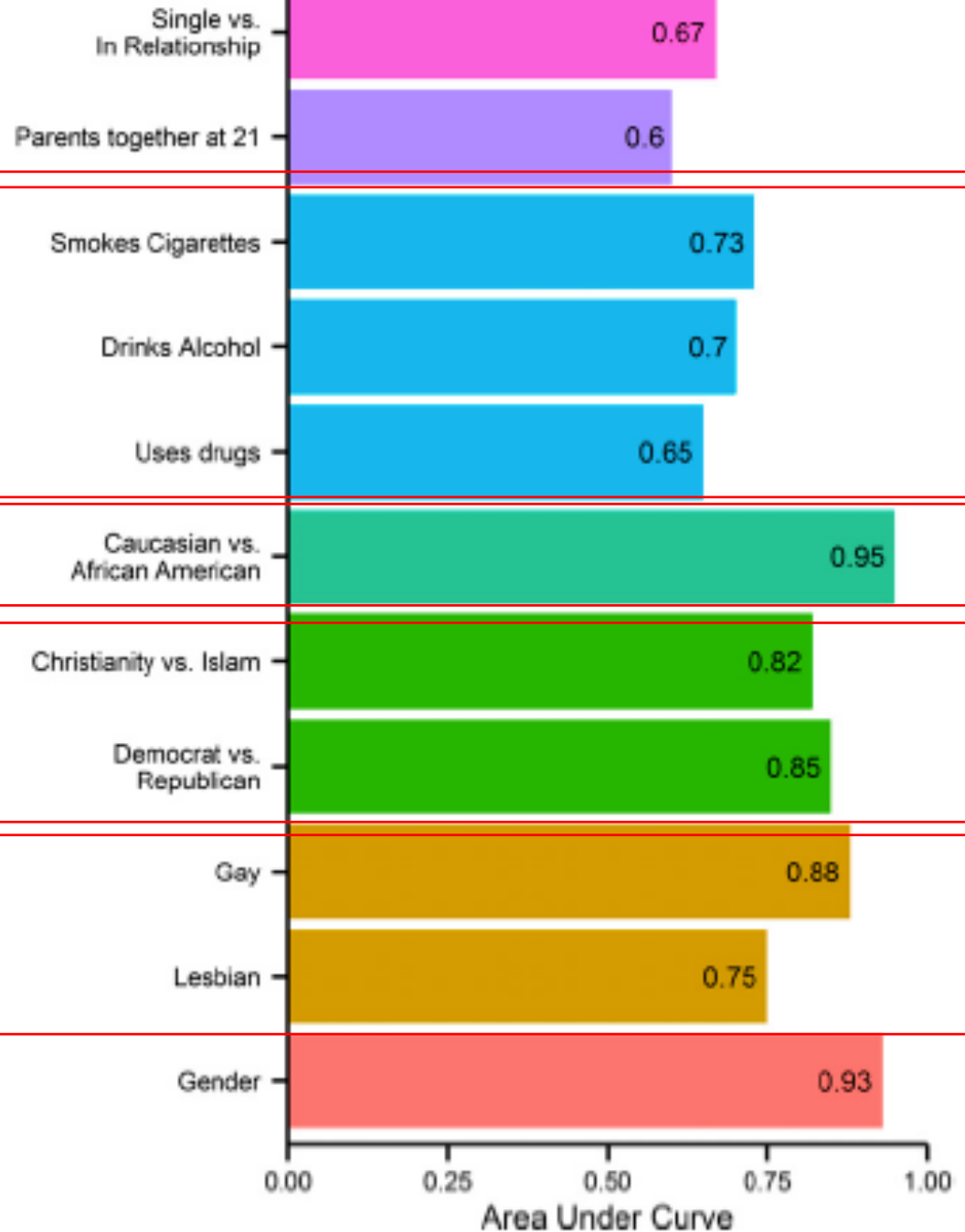
Shocking outcomes

Private habits
with relationship
with health

Racial
group

Religion
and political
thinking

Sexual
habits



INPS data breach: analysis of an italian case

   **Norme & Tributi** Lavoro e previdenza

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31 marzo 2020

Coronavirus
INPS
Iva
Pasquale Tridico
Cme

 Salva  27 Commenta

   ...

SERVIZIO | PARTITE IVA E AUTONOMI

Bonus 600 euro, Tridico: «Abbiamo chiuso il sito dell'Inps, segnalati attacchi hacker»

L'Inps ha chiuso per diverse ore il proprio sito, dopo il boom di accessi e gli attacchi hacker subiti. Proprio il 1° aprile si era aperto il canale telematico per inviare le domande per il bonus da 600 euro riservato a 5 categorie: liberi professionisti e co.co.co; autonomi iscritti alle gestioni speciali; stagionali del turismo; lavoratori agricoli; addetti dello spettacolo

di Francesca Barbieri

   **Tecnologia** Economia digitale

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1 aprile 2020

INPS
Pasquale Tridico
garante per la privacy
Antonello Soro
Francesco V.

 Salva  25 Commenta

   ...

CYBERATTACCO O PROBLEMI DEL PORTALE?

Sito Inps, perché l'ipotesi dell'attacco hacker non convince

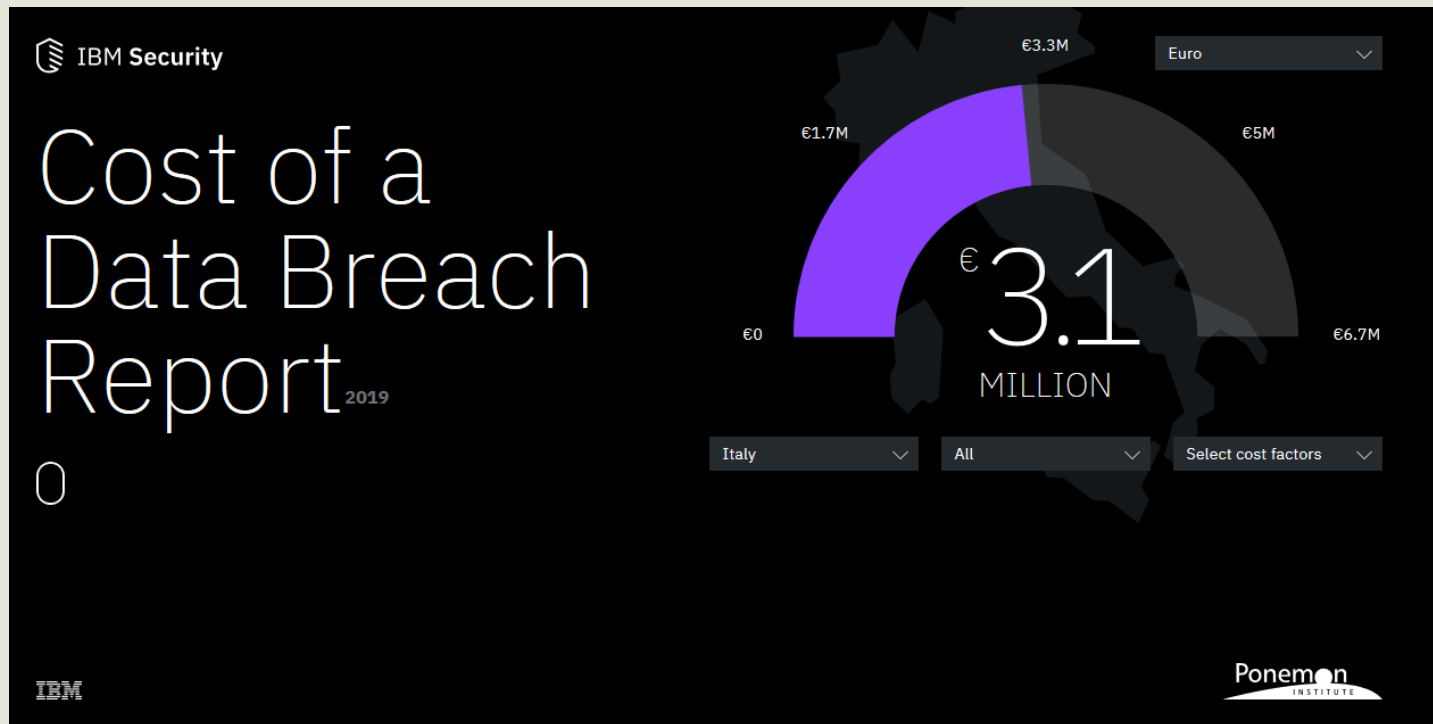
Il servizio ha funzionato per la prima parte della giornata, poi si è evidenziato un data breach che ha provocato il caos. Le ipotesi possibili

di Biagio Simonetta



(ANSA)

Cost of a data breach



Cost for citizens



Current estimates:
\$ 1000 – 3000 per year



Which is the real cost?

«Emerge un legame profondo tra libertà, dignità e privacy, che ci impone di guardare a quest'ultima al di là della sua storica definizione come diritto ad essere lasciato solo.»

«Quale dignità rimane ad una persona divenuta prigioniera di un passato interamente in mani altrui, di cui deve rassegnarsi ad essere espropriato?»

Stefano Rodotà - Privacy, libertà, dignità

